

QuantumDynamics Framework and Capability Notes

Paul A. Brown

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Abstract

QuantumDynamics is a one-dimensional, two-diabatic-state wavepacket framework for hydrogen-atom transfer between coupled potential wells. It combines split-operator propagation, optional Markovian or exponentially correlated Langevin centroid damping, optional stochastic modulation of either well, MPI-distributed trajectories, OpenMP/FFTW acceleration, and physical-unit input/output. This note records current equations, controls, tested behavior, analysis capability, known limitations, and development status.

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1 Scope and current status

Current code is suitable for reproducible exploratory dynamics and ensemble kinetics. Input validation, unit conversion, FFT ordering, random-stream portability, HDF5 failure handling, state-selective damping, and output safety have regression coverage. Both HDF5-enabled and HDF5-disabled builds are tested.

Code models one nuclear coordinate and two coupled diabatic amplitudes. It is not a general multidimensional quantum chemistry package, a density-matrix solver, or a first-principles electronic-structure engine.

2 Dynamical model

2.1 Two-state Hamiltonian

Wavefunction is

$$\Psi(x, t) = \begin{pmatrix} \psi_1(x, t) \\ \psi_2(x, t) \end{pmatrix}, \quad (1)$$

with Hamiltonian

$$\hat{H} = -\frac{1}{2m} \frac{\partial^2}{\partial x^2} \mathbf{I} + \begin{pmatrix} V_{11}(x, t) & V_{12}(x) \\ V_{12}(x) & V_{22}(x, t) \end{pmatrix}. \quad (2)$$

Base diabatic wells are

$$V_{11}(x) = \frac{1}{2}k_1(x - x_1)^2 + c_{4,1}(x - x_1)^4 + \Delta_1, \quad (3)$$

$$V_{22}(x) = \frac{1}{2}k_2(x - x_2)^2 + c_{4,2}(x - x_2)^4 + \Delta_2. \quad (4)$$

Quartic terms are enabled by anharmonic model selections. Gaussian coupling is

$$V_{12}(x) = V_c \exp \left[-\frac{(x - x_c)^2}{2\sigma_c^2} \right], \quad (5)$$

with an exponential alternative. Lower and upper adiabatic surfaces are the eigenvalues of the local 2×2 diabatic potential matrix.

2.2 Split propagation

One Hamiltonian step uses symmetric potential splitting,

$$e^{-i\hat{H}\Delta t} \approx e^{-i\hat{V}\Delta t/2} e^{-i\hat{T}\Delta t} e^{-i\hat{V}\Delta t/2}. \quad (6)$$

Local potential exponentials use an analytic Hermitian 2×2 form. Kinetic propagation uses forward and inverse FFTW transforms. Dedicated plans are created for both diabatic components. Even and odd grid sizes have tested FFT frequency ordering.

2.3 Momentum Langevin bath

Markovian damping uses an exact finite-step Ornstein–Uhlenbeck centroid update consistent with classical fluctuation–dissipation statistics. Colored damping uses an exponential memory kernel,

$$\gamma(t) = \frac{\gamma}{\tau_c} e^{-t/\tau_c}, \quad \tau_c = \frac{2}{\text{FWHM}}, \quad (7)$$

with auxiliary memory and stationary random-force variables. The controls `damp_reactant` and `damp_product` select the diabatic component that receives the momentum translation. Enabling both recovers shared-centroid damping.

This thermostat is a classical-FDT pure-state construction. The QGLE label does not imply a full quantum-frequency-dependent fluctuation–dissipation kernel.

2.4 Stochastic potential bath

Either diabatic well can receive coordinate-shift, curvature, or enthalpy modulation. Noise can be white or Ornstein–Uhlenbeck colored and can be shared or independent between wells. Colored and white processes begin with their stationary Gaussian marginal distribution. Mode-dependent input amplitudes use length, force-constant, or energy units as appropriate.

2.5 Absorbing boundaries

Optional smooth edge masks suppress periodic FFT wraparound. Within an edge layer of width w , each amplitude is multiplied after a full time step by

$$M(x) = \exp \left[-\Gamma_{\text{abs}} \Delta t \left(\frac{d_{\text{edge}}(x)}{w} \right)^p \right]. \quad (8)$$

Absorbed norm represents outgoing flux. Norm-conservation tests apply only when edge absorption is disabled.

3 Units and input boundary

Namelist input uses chemistry-facing units and converts once to atomic units. Output writers convert back to physical units.

Quantity	Input/output unit	Internal unit
Coordinate, widths	angstrom	bohr
Time	fs	atomic time
Energy and coupling	cm ⁻¹	hartree
Force constant	cm ⁻¹ angstrom ⁻²	hartree bohr ⁻²
Quartic coefficient	cm ⁻¹ angstrom ⁻⁴	hartree bohr ⁻⁴
Rates and FWHM	fs ⁻¹	atomic-time ⁻¹
Mass	u/amu	electron mass
Temperature	K	inverse hartree through β
Wave number	angstrom ⁻¹	bohr ⁻¹

4 Primary controls

Control	Meaning
<code>nx</code> , <code>xmin</code> , <code>xmax</code>	Periodic real-space grid and physical box.

Control	Meaning
<code>dt, nsteps, save_every</code>	Physical time step, total steps, and output cadence.
<code>ntraj, seed0</code>	Number of independent stochastic trajectories and base seed.
<code>temperature_k, mass_amu</code>	Thermodynamic temperature and transferred-particle mass.
<code>gamma, use_colored, kernel, fwhm</code>	Momentum bath strength and memory kernel.
<code>damp_reactant, damp_product</code>	Reactant/product diabatic damping switches.
<code>pot_model, k1, k2, x1, x2</code>	Well model, stiffnesses, and minima.
<code>v1_shift, v2_shift, c4_1, c4_2</code>	Energy offsets and quartic coefficients.
<code>v12, sigma, want_coupling</code>	Diabatic coupling magnitude, width, and switch.
<code>x0, p0, sigma0</code>	Initial Gaussian packet center, wave number, and width.
<code>bath_pot_*</code>	Stochastic potential mode, amplitude, color, and well selection.
<code>use_absorber</code>	Enable smooth edge absorption.
<code>absorber_width</code>	Edge-layer width in angstrom.
<code>absorber_rate</code>	Maximum mask rate in fs^{-1} .
<code>absorber_power</code>	Polynomial smoothness exponent.
<code>hdf5, out_prefix</code>	Output format and filename prefix.
<code>write_initial</code>	Write exact $t = 0$ snapshot before propagation.

Input may be supplied as `./qle_1d INPUT.nml`. Running `./qle_1d -help` cannot start a trajectory or overwrite output.

5 Parallel execution and reproducibility

MPI distributes complete trajectories; OpenMP and threaded FFTW accelerate each trajectory. Random streams depend on trajectory number rather than MPI rank, so changing rank count does not change a trajectory’s stochastic sequence. Output filenames retain rank identifiers, so filenames can change when MPI layout changes even when numerical streams do not.

6 Output and analysis

HDF5 stores physical coordinates, times, wavefunctions, populations, position expectations, and potential surfaces. Static potential surfaces are stored only in the first frame; stochastic surfaces remain frame-resolved. HDF5 operations fail loudly rather than silently leaving incomplete files. ASCII mode writes snapshot densities, observable time series, and static or frame-resolved PES files.

`ShowDyn.py` reads trajectory files, displays HDF5 trees, plots snapshots, animates densities and surfaces, and plots reactant/product probabilities. `analyze_ensemble.py` combines rank/trajectory files, checks common time grids, writes mean and standard-error populations, performs bounded mono- and biexponential fits, and can bootstrap trajectories for empirical rate intervals.

7 Verification status

Regression coverage includes:

- physical-unit round trips for all supported dimensions;
- zero-damping norm conservation;
- analytic white and colored centroid relaxation;
- fluctuation–dissipation ensemble means and variances;
- product-only damping isolation;
- stationary stochastic-potential initialization;
- coupling and anharmonic-potential switches;
- even/odd FFT frequency ordering and FFTW plan creation;
- smooth absorber norm removal;
- safe command-line help and explicit input paths;
- exact $t = 0$, ASCII, dynamic-PES, and HDF5 output;
- HDF5 and non-HDF5 builds under strict compiler warnings.

8 Example two-timescale kinetics

A 1.5 ps product-damped trajectory at 10 K produced a bounded biexponential product-population fit with representative rates

$$k_{\text{fast}} \simeq 6.22 \times 10^{-2} \text{ fs}^{-1}, \quad (9)$$

$$k_{\text{slow}} \simeq 2.47 \times 10^{-3} \text{ fs}^{-1}. \quad (10)$$

The fit strongly improved AIC relative to a monoexponential model. Fast and slow components are consistent with prompt/over-barrier and residual tunneling channels, respectively, but this assignment is not unique from one stochastic pure-state trajectory. Publication claims require ensemble statistics and mechanism-control calculations.

9 Limitations and update status

Item	Status	Note
Missing $t = 0$ output	Implemented	Exact initial frame now optional and enabled by default.
Periodic boundary return	Implemented	Smooth absorbing layers available; convergence remains user responsibility.
MPI/rank aggregation	Implemented	Ensemble tool reports means, SEM, fits, and bootstrap intervals.
Repeated static HDF5 PES	Implemented	Static PES stored once; stochastic PES remains time resolved.
Single-trajectory rate uncertainty	Implemented	Ensemble bootstrap supported when multiple trajectories exist.
Checkpoint/restart	Planned	Needs versioned serialization of wavefunctions, RNG, Langevin memory, potential-bath state, time, and output indices.

Item	Status	Note
Single-file parallel HDF5	Planned	Current design intentionally uses independent per-rank files.
Full quantum FDT	Planned	Current thermostat uses classical centroid covariance.
Density-matrix decoherence	Planned	Current evolution uses pure-state trajectories and phenomenological kicks.
Multidimensional transfer	Planned	Current grid contains one nuclear coordinate.
Automated convergence campaign	Planned	Users must sweep Δt , box, grid, absorber, damping, and trajectory count.
Mechanism-proof rate assignment	Planned	Requires energy/barrier, coupling-off, isotope, and ensemble controls.

Checkpoint/restart is deliberately not added as an unversioned binary dump. Incorrect restoration of random and memory variables would produce apparently valid but non-reproducible trajectories. A future implementation should use a versioned HDF5 checkpoint schema and exact continuation tests.

10 Recommended production protocol

1. Run `make test` for the selected compiler and library stack.
2. Perform time-step, grid-spacing, box-size, and absorber convergence.
3. Use several independent trajectories and aggregate with `analyze_ensemble.py`.
4. Monitor total norm, absorbed norm, edge density, and high-frequency FFT content.
5. Compare no-damping, reactant-only, product-only, and both-state damping.
6. Establish mechanism with barrier-energy and coupling controls before labeling fitted rates as over-barrier or tunneling.
7. Archive input, executable revision, dependency versions, seeds, raw data, and analysis outputs.

11 Build and run summary

```
make
make test
./qle_1d INPUT.nml
python analyze_ensemble.py 'run.traj*.rank*.h5' --output ensemble --bootstrap 500
python ShowDyn.py run.traj000001.rank0.h5 --populations --no-usetex
```

HDF5 may be disabled with `make USE_HDF5=0`. FFTW, MPI, OpenMP, and HDF5 prefixes remain configurable through the build environment.